

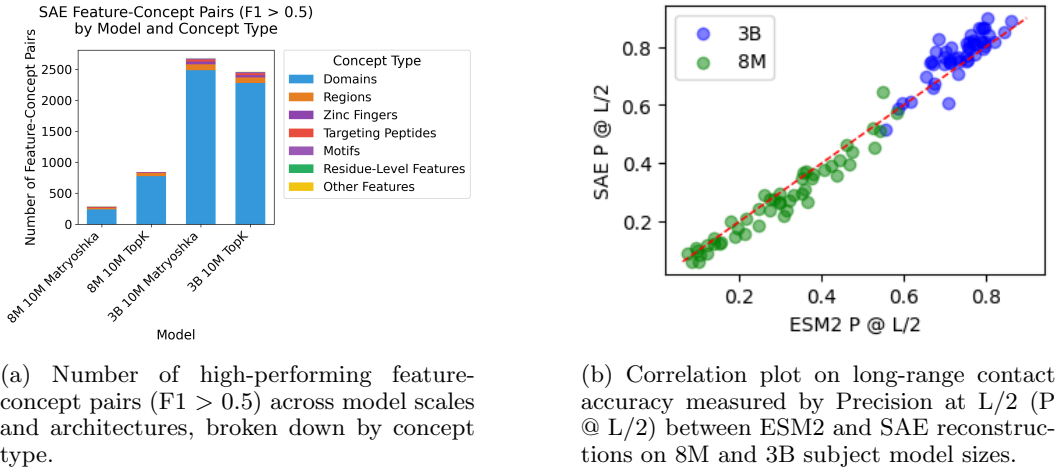


Figure 3: Analysis of feature-concept relationships and long-range contact accuracy.

4 FURTHER EVALUATIONS

4.1 SWISS-PROT CONCEPT DISCOVERY

We evaluate feature interpretability through alignment with Swiss-Prot annotations following Simon and Zou (2024)’s methodology (details in Appendix E.1), analyzing 30,871,402 amino acid tokens across 476 biological concepts. Features capturing concepts ($F1 \geq 0.5$) are identified using domain-level recall with post-hoc $[0,1]$ activation normalization across architectural variants.

Comparing ESM2-3B versus ESM2-8M models (dictionary size 20,480, sparsity $k=100$), 3B variants substantially outperform their 8M counterparts. The 3B Matryoshka and TopK models identify 233 concepts (48.9%) with $F1 > 0.5$, generating 2,677 and 2,461 high-quality feature-concept pairs respectively. In contrast, 8M Matryoshka and TopK variants capture only 72 (15.1%) and 95 (20.0%) concepts, with 287 and 844 pairs respectively. Performance gains are particularly pronounced in protein domains (76% versus 19.5-28.1% coverage). Head-to-head comparisons show 3B models achieve higher F1 scores on over 400 concepts (mean improvement 0.25), while architectural differences within each scale remain minimal (Figure 6, details in Appendix E.2).

4.2 CONTACT MAP PREDICTION

Background. Following Zhang et al. (2024), we evaluate our SAEs’ ability to capture coevolutionary statistics through contact map prediction using the Categorical Jacobian. Details in Appendix G. This provides an unsupervised test of whether our compressed representations preserve the structural information encoded in the original model.

Results. We assess contact prediction accuracy using the precision at $L/2$ metric ($P @ L/2$), which measures the fraction of correctly predicted contacts among the top $L/2$ predicted long-range contacts, where L is the protein sequence length. Figure 3b shows the correlation between contact prediction accuracy of ESM2 and our SAE reconstructions across different model scales. The 3B model demonstrates consistently higher precision compared to the 8M model, with reconstructions closely tracking the original ESM2 predictions. This suggests that larger language models capture more robust coevolutionary signals that can be effectively compressed by our approach.

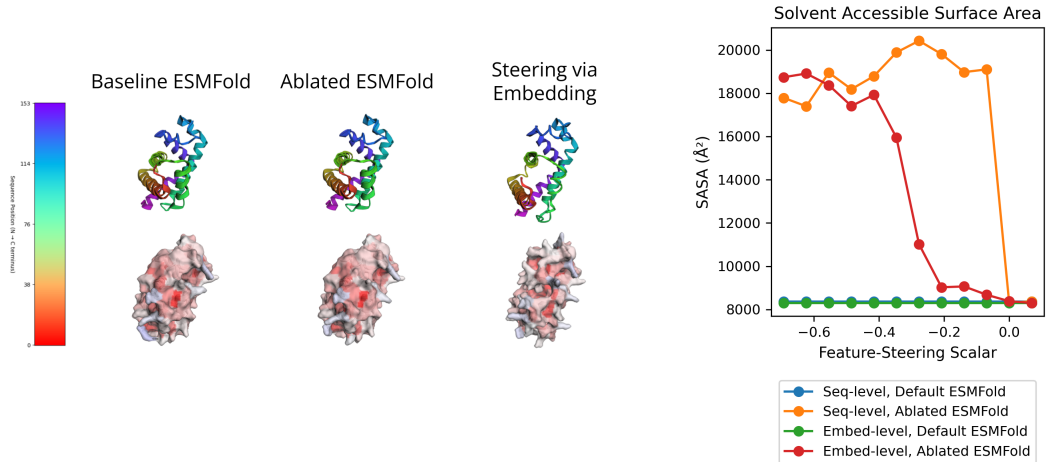
5 CASE STUDY: SAE FEATURE STEERING ON ESMFOLD

5.1 SAEs ON ESMFOLD HELP IDENTIFY STRUCTURAL FEATURES

Building on previous work by Simon and Zou (2024), we investigated whether targeted feature manipulation of ESM2-3B could induce coordinated changes in both sequence composition and predicted structure through ESMFold’s pipeline. Using our Matryoshka SAE trained on ESM2-3B, we identified a feature strongly correlated with residue hydrophobicity. Following the feature steering methodology of Templeton et al. (2024) and detailed in Appendix Section H, we steer this feature ($\alpha = -0.275$) and observe significant changes in predicted protein surface accessibility of myoglobin (PDB ID: 1MBN) while maintaining structural integrity (Figure 4a).

Steering increased total Solvent Accessible Surface Area (SASA) by 31.5% (from 8,369.5 to 11,009.3 Å²) with minimal structural disruption (RMSD 2.76 Å) in the expected range of Figure 2b. Feature intervention affected both sequence-level predictions via steering toward more hydrophilic residues and direct structural predictions of ESMFold (Figure 9). Critically, steering the hidden representation at layer 36 alone was sufficient to induce significant structural changes consistent with increased hydrophilicity, even when providing ESMFold with the correct input sequence (Figure 4b). Further details found in Appendix Section H.2 and feature validation in Appendix Section H.3.

6 DISCUSSION



(a) Structural visualization of feature steering effect on myoglobin with $\alpha = -0.275$ on selected feature. Bottom row surface representation is colored by computed SASA, with blue as low, white as medium, and red as higher SASA.

(b) Solvent accessible surface area (SASA) changes under different steering conditions.

Figure 4: Feature steering and SASA analysis.

This work advances PLM interpretability by scaling SAEs to ESM2-3B, extending recent work using SAEs to interpret PLMs to the structure prediction task (Simon and Zou (2024); Adams et al. (2025)). Through a combination of increasing subject model size, leveraging the Matryoshka architecture, and targeted interventions on ESMFold’s structure predictions, we present the first work applying mechanistic interpretability to protein structure prediction. A key limitation is that our focus is exclusively on how PLM-derived sequence representations inform structure prediction. Looking ahead, we aim to establish formal connections between SAE sequence representations and equivariant representations of geometric data. Additional limitations and future directions can be found in Appendix B and C, respectively.

7 APPENDIX

A RELATED WORK

Recent work has focused on mechanistically interpreting protein language models (PLMs) to understand how they achieve remarkable success in protein modeling and design. Early interpretability studies examined PLM internals, with Vig et al. (2021) and Rao et al. (2021) discovering that attention patterns encode structural relationships between amino acids. Later work extended these findings, showing attention could identify functionally important regions like allosteric sites (Kannan et al. (2024); Dong et al. (2024)).

Building on this attention-based analysis, Zhang et al. (2024) provided key mechanistic insights by showing that PLMs primarily learn by storing and looking up coevolutionary patterns preserved through evolution, rather than learning fundamental protein physics. By analyzing what sequence features influence contact predictions through a "categorical Jacobian" method, they demonstrated that PLMs memorize statistics of co-evolving residues analogous to classic evolutionary models.

Sparse autoencoders (SAEs) have emerged as a powerful new tool for interpreting these models. Simon and Zou (2024) developed InterPLM, extracting thousands of interpretable features from ESM-2 that correspond to known biological concepts like binding sites, structural motifs, and functional domains. Their work revealed that most biological concepts exist in superposition within the model’s neurons, as SAEs vastly outperformed individual neurons at capturing these concepts. Concurrently, Adams et al. (2025) explored different SAE training approaches and evaluated their biological relevance through novel downstream tasks. In this work, we scale SAE training and evaluation to ESM2-3B and introduce Matryoshka SAEs (Bussmann et al. (2024); Nabeshima (2024)) with hierarchical encoding to steer ESMFold’s structure prediction capabilities for the first time.

B ADDITIONAL LIMITATIONS

Below we outline additional limitations of our work. First, as mentioned earlier, although we take an important step toward mechanistic interpretability of protein structure prediction, our focus is exclusively on how PLM-derived sequence representations inform structure prediction. The interpretability of ESMFold’s structure prediction head is left for future investigation. Second, our interventions on ESMFold were restricted to single-feature manipulations on embeddings from individual layers rather than exploring combinations of features or cross-layer interactions. Third, our analysis is limited to the 8M and 3B models of ESM2; evaluating a broader range of model sizes could reveal whether SAE performance eventually plateaus. Fourth, ESMFold is no longer state-of-the-art compared to newer diffusion-based protein structure prediction methods Abramson et al. (2024); Wohlgend et al. (2024); Watson et al. (2023).

Fifth, recent work suggests that SAEs can learn different features even when trained on the same data, implying that our results may be sensitive to both initialization and the specific UniRef50 sample used (Paulo and Belrose (2025)). Sixth, our structural steering experiments focused primarily on surface accessibility, leaving other potential structural properties unexplored, and the observed hydrophobicity effects might be achievable through simpler approaches such as steering smaller PLMs or using supervised linear probes. Seventh, for Matryoshka SAEs, we do not report how the number of groups and group size choices impact results. Finally, although our ablation studies demonstrate preservation of structural information, we have not fully characterized how interactions among SAE features might affect ESMFold’s folding mechanism. We anticipate that future work leveraging our open-source models developed with substantial computational resources will address these limitations and yield further insights for the community.